TECHNICAL DOCUMENTATION

RAG-Based Customer Support Assistant using LangGraph & HITL

---

1. Introduction

◆ What is RAG (Retrieval-Augmented Generation)?

Retrieval-Augmented Generation (RAG) is an AI architecture that combines:

- **Information Retrieval** (searching relevant data from documents)
- **Text Generation** (using a language model to generate responses)

Working:

User Query → Retrieve relevant chunks → Pass to LLM → Generate Answer

---

◆ Why RAG is Needed

Traditional LLMs:

- Do not have real-time knowledge
- Can generate hallucinated responses

RAG solves this by:

- Using external knowledge sources
- Improving accuracy and reliability

◆ Use Case Overview

This project implements a **Customer Support Assistant** that:

- Answers user queries from a PDF knowledge base
- Provides contextual responses
- Escalates complex queries to a human agent

---

2. System Architecture Explanation

The system consists of the following components:

◆ Document Pipeline

- Loads PDF

- Splits into chunks
- Converts into embeddings

◆ **Vector Database**

- Stores embeddings
- Enables similarity-based retrieval

◆ **Retrieval Layer**

- Fetches relevant chunks for a query

◆ **LLM Layer**

- Generates answers using retrieved context

◆ **Workflow Layer (LangGraph)**

- Controls execution flow
- Handles routing decisions

◆ **HITL System**

- Handles escalations
- Ensures reliability

---

3. Design Decisions

◆ **Chunk Size = 500**

- Balances context and performance

◆ **Chunk Overlap = 50**

- Prevents loss of important context

◆ **Embedding Model**

- HuggingFace (all-MiniLM-L6-v2)

- Lightweight and efficient

◆ **Vector Database**

- ChromaDB chosen for:
  - Simplicity
  - Speed
  - Local storage

◆ **Retrieval Strategy**

- Semantic similarity search
- Ensures relevant context

◆ **Prompt Design**

- Combines context + query
- Ensures accurate answers

---

4. Workflow Explanation (LangGraph)

The system uses a **graph-based workflow**:

Input → Process Node → Decision → Output / HITL

◆ **Node Responsibilities**

✔ **Process Node**

- Performs retrieval
- Generates response
- Decides escalation

✔ **HITL Node**

- Accepts human input
- Overrides AI response

✔ **Output Node**

- Returns final answer

🔹 **State Transitions**

Question → Context → Answer → Escalation Flag

---

5. Conditional Logic

🔹 **Intent Detection**

The system determines whether:

- Query is answerable
- Or requires escalation

🔹 **Routing Decisions**

✔ **Answer Directly When:**

- Context is available
- Response is meaningful

✔ **Escalate When:**

- No relevant data found
- Low confidence response
- Complex query

---

6. HITL Implementation

🔹 **Role of Human Intervention**

- Handles edge cases
- Improves reliability
- Prevents incorrect answers

🔹 **Flow**

Query → AI Processing → Escalation → Human Response → Output

🔹 Benefits

- Increased accuracy
- Better user experience

🔹 Limitations

- Slower response time
- Requires human availability

---

7. Challenges & Trade-offs

🔹 Retrieval Accuracy vs Speed

- Larger chunks → better context
- Smaller chunks → faster retrieval

🔹 Chunk Size vs Context Quality

- Too large → inefficient
- Too small → loss of meaning

🔹 Cost vs Performance

- Advanced models → high cost
- Lightweight models → lower accuracy

---

8. Testing Strategy

🔹 Testing Approach

- Functional testing
- Edge case testing
- HITL validation

🔹 Sample Queries

Query	Expected Behavior
Refund policy	Direct answer
Shipping time	Direct answer
Unknown query	HITL triggered

🔹 Validation

- Check retrieval accuracy
- Verify routing logic
- Confirm HITL activation

---

9. Future Enhancements

🔹 Multi-Document Support

- Handle multiple PDFs

🔹 Feedback Loop

- Improve system over time

🔹 Memory Integration

- Maintain conversation history

🔹 Deployment

- Web app (Streamlit / FastAPI)
- Cloud deployment

---

10. Conclusion

This project demonstrates a complete **RAG-based AI system** with:

- Efficient document retrieval
- Context-aware response generation
- Workflow orchestration using LangGraph

- Human-in-the-Loop integration

It provides a scalable and practical solution for real-world customer support automation.